Unit 11, Lesson 6: Writing Exponential Functions From a Written Description

Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

Learning Targets

- ☐ I can write an exponential function from a written description.
- ☐ I can classify a real-world context modeled by an exponential function as exponential growth or decay.

11.6.1 Warm-Up: Math Talk

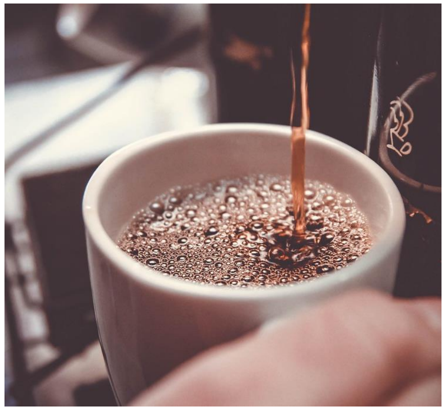
The map shows the Southeastern United States to scale. Using the map, fill in the blanks below. Be prepared to share your reasoning with the class.

1. Florida has an area that is about _________ times bigger than South Carolina.

2. The area of Florida is $65{,}755$ square miles. Based on your estimate, the area of South Carolina is ___________ square miles.

11.6.2 Guided Instruction: Writing Exponential Functions

1. A typical cup of coffee contains about 100 milligrams (mg) of caffeine. Every hour approximately 6% of the amount of caffeine in the body is metabolized and eliminated.



   a. Discuss with your partner whether the situation represents exponential growth or exponential decay. Justify your choice.

   b. What is the rate of growth or the rate of decay?

   c. What is the initial amount of caffeine consumed in this scenario?

   d. Let C represent the amount of caffeine in the body, in mg, and t represent the number of hours since a cup of coffee was consumed. Write C as a function of t.

2. In 2020, the population of Florida was estimated to be 21.7 million people, with an annual rate of increase of 1.12%.

   a. Explain whether the population can be modeled by exponential growth or exponential decay.

   b. Wyatt and Carter both wrote a function to model Florida's population growth, where x is the number of years since 2020 and P is the population in millions. Both students made a mistake. Circle the error each student made.

Wyatt	Carter
$P(x) = 21.7(1.12)^x$	$P(x) = 21.7(1 + 0.112)^x$

   c. Discuss with your partner the errors Wyatt and Carter made.

   d. Write the correct function that models the context using both forms.

$\boldsymbol{P(x) = a(b)^x}$	$\boldsymbol{P(x) = a(1+r)^x}$

11.6.3 Try It: Writing Exponential Functions

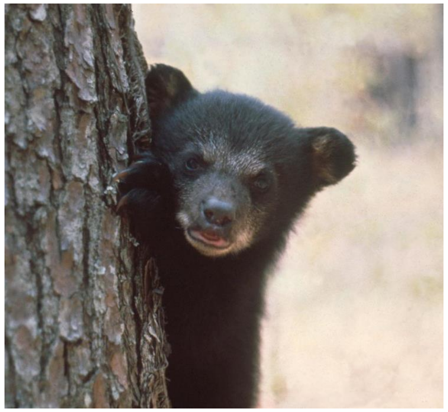
Work independently to write the function for the two scenarios.

1. The value of a 2021 Ford Bronco was $\$44{,}205$ when it went on the market. It depreciates at a rate of 10.5% per year.

   a. Is this an example of growth or decay?

   b. Write a function that models the car's value, where V is the value of car and t is the number of years since 2021.

2. The Florida Fish and Wildlife Conservation Commission have found that the Florida black bear population is increasing due to the Florida Black Bear Management Plan. In 2019, there were $4{,}050$ bears in Florida with a rate of increase of 6.5% per year.



   a. Is this an example of growth or decay?

   b. Write an equation that models the population, where B is the number of black bears and x is the years since 2019.

3. Compare your answers to questions 1-2 with your partner and if necessary, resolve any differences.

11.6.4 Collaboration: Exponential Equations in Real-World Contexts

U.S. lawmakers have tried to pass legislation that would reduce the consumption of salt. Versions of this law have been proposed in the past that would mandate restaurants to decrease sodium levels in the food they serve by an exponential rate. The sodium level in food is the amount of salt in the food.

It is estimated that all U.S. restaurants use a collective total of 5 million grams of salt a year. If the law were to pass, they would be required to reduce to 4.875 million grams the following year and continue reducing at a constant percentage each year until a goal is reached.

1. Discuss with your partner how the values given in this scenario can be used to write an exponential function that models the situation.

2. Two students discussed how to determine the value of b to write an exponential function in the form $y = a(b)^x$. Their thinking is shown in the table.

Jayden	Ella
"I would subtract 4.875 million from 5 million to find the rate of change from one year to the next."	"I would divide 4.875 million by 5 million to find the common ratio from one year to the next."

Explain which student is correct.

3. Find the value of b in this scenario.

4. Discuss with your partner which value in the scenario represents a.

5. Work with your partner to write the function rules in the forms indicated, where S is the total amount of salt that can be used, in millions of grams, for the given years, x, since the law passed.

$\boldsymbol{S(x) = a(b)^x}$	$\boldsymbol{S(x) = a(1 \pm r)^x}$

6. What is the constant percent rate of change by which the law would require restaurants to reduce the amount of salt served in their food each year?

11.6.5 Your Turn!

1. In 1987, there were 17 Starbucks locations. The number of shops has increased 29% annually since then.

   Write the function that models the number of Starbucks since 1987. Let t be the number of years since 1987 and S be the number of shops.

2. There were 1.462 million paid subscriptions for streaming music in 2010. Since then, there has been a 54.4% per year increase in paid music streaming subscriptions.

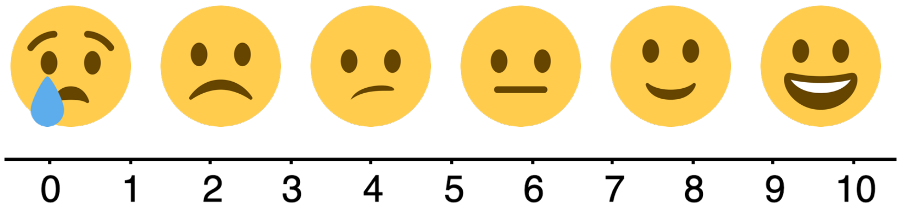
   Write a function that represents the number of paid subscriptions since 2010. Let x be the years since 2010 and P be the number of paid music streaming subscriptions, in millions.

3. An adult takes 400 milligrams (mg) of ibuprofen. Each hour, the amount of ibuprofen in the person's system decreases by 29%.

   Write a function that represents the amount of ibuprofen, A, in mg, after h hours.

11.6.6 Wrap-Up: Reflection

A scale is shown with emojis that correlate to different levels of confidence.



1. Draw the emoji that best describes your understanding of each learning target from this unit.

Emoji	Learning Target
	I can classify a function as representing exponential growth or decay from a graph or table.
	I can determine the key features of an exponential function from a table, graph, or exponential function.
	I can graph an exponential function given the equation.
	I can graph an exponential function given a written description and determine its key features.
	I can write an exponential function from a graph or table.
	I can write an exponential function from a written description.

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